

Behavioral Risks and Cybersecurity Vulnerabilities: Memo for CEO

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Abstract

As manufacturing organizations expand through remote work at an all-time high, the reliance on employee technology use introduces never-seen-before vulnerabilities. This report examines how human behavior and personnel turnover influence the frequency of security breaches. Common errors and industry data provide the necessity for a structured cybersecurity plan.

The Human Element

The current posture of our organization is at immediate risk because of how our employees interact with technology. While many people believe that cyber defense is only the responsibility of the technical people, research indicates that human choices currently occurring in our organization are a primary cause of security failures. When employees are not warned of the dangers, both effectively and actionably, they can be easily victimized by the external criminals who want to steal company data or money. Global costs of cybercrime reached nearly \$1 trillion USD in 2020, which is a massive increase of more than 50% since 2018. This means that the danger is growing for all businesses, so businesses need a much better way to manage risks (Cremer, et al., 2022).

According to Schoenfield (2019), different kinds of attackers employ a variety of techniques to succeed in their ploy. In our manufacturing industry, there is an extremely high chance that these agents will utilize social engineering against our employees.

Most of the employees on our marketing team use social media to market the latest manufactured products and find new hires, which makes them visible and susceptible to hackers.

Attackers often use social engineering to trick employees into doing something wrong by *pretending* to be a boss or friend. If an employee is swindled by a fake message, they accidentally allow a hacker to enter our private systems. Because 300 of our staff work in different offices and states, they do not have the same support as a physical office, making them easier targets for these tricks. Without a solid and organized plan, we are leaving our front-line workers open to being tormented by these evolving methods.

The Risks of Personnel Turnover

The high rate of turnover in our company creates specific gaps that attackers like to use as leverage. Each time a person leaves, there is the risk that their access to our data is not removed quickly enough. In the manufacturing sector, these types of vulnerabilities can lead to business interruption and detrimental financial losses. If a former worker is victimized by a hacker who steals their old password, that hacker can enter our network without being noticed (Cremer, et al., 2022).

Furthermore, new employees join the company without knowing our security expectations. Because there is no set cyber plan, these new workers might use simple passwords or click on malicious links. High turnover means we are constantly teaching new people, and any delay in this teaching makes us more likely to be victimized. A new door opens for a potential attack every time a new person starts. We must include security training as a permanent part of our company culture.

Data Analysis of Cybercrime Trends

To understand the unique threats we face, we must look at how different types of crime are changing. Manufacturing is the fourth most targeted industry, with 300 compromises reported

in the past year (Identity Theft Resource Center, 2026, p. 6). The following table presents information on typical cybercrime problems that are often exacerbated by our company's two biggest growing attributes, high turnover and remote access.

Attack Category	Sub-Category / Vector	Total Compromises	Total Victim Notices
Cyberattacks	Total	2,656	227,010,931
	Phishing / Smishing / BEC	466	21,540,812
	Ransomware	143	12,855,304
	Credential Stuffing	28	5,642,881
	Malware	25	2,125,441
	Not Specified (Vague)	1,948	184,846,493
System & Human Error	Total	230	6,220,273
	Correspondence (Email/Letter)	115	3,115,442
	Cloud Security Failure	24	1,224,335
	Misconfigured Firewall	8	442,125
	Lost Devices / Documents	4	245,612
Physical Attacks	Total	66	23,486
	Skimming Devices	34	12,443
	Document / Device Theft	19	8,432

Adapted from Identity Theft Resource Center. (2026). 2025 Annual Data Breach Report, p. 27.

Attackers primarily target our employees, demonstrating that a company with a high employee turnover rate and a sizable remote workforce is much more likely to suffer from human-centered vulnerabilities than from technical malfunctions alone.

Case Studies of Infrastructure Impact

The danger to our manufacturing company is like what other large industries have faced. For instance, a 5,500-mile system was rendered inoperable by a ransomware attack on the

Colonial Pipeline. Because 45% of the U.S. East Coast was unable to obtain the necessary fuel, this strike had a negative economic impact.

Another example of how a company can be taken advantage of is the NotPetya attack in 2017. This attack spread on its own through computer networks and caused \$10 billion USD in damage worldwide. Events like these show that attackers do not just steal files; they stop the work of the company entirely. For a company like ours with 500 employees, the cost of being victimized in this way would be very hard to recover from.

Remote Work and Global Vulnerability

Our 300 remote workers are at a higher risk because they often use networks that we do not control. During the COVID-19 pandemic, attacks on people working at home increased significantly because they were more vulnerable. When working alone, employees are more likely to make a mistake and become victims of ransomware.

Since they travel to advertise our products, our staff bring company phones and laptops into public spaces. If an employee misplaces a device or uses the wrong internet connection, they could have their data compromised by a bad actor immediately. We must ensure that our remote workers know what precautions to take when they are not in the office.

The Problem of Data Gaps

Many companies that have been hacked choose not to reveal that anything negative happened to them. This makes it difficult for researchers and security specialists to create more efficient defense measures. When a business reputation is damaged, it can be difficult to determine the insurance rates or restitution because the costs are unclear with current data. We

must admit that being unaware of our risk due to underreporting is a major problem (Cremer et al., 2022).

Conclusion

Bad actors most frequently attack our organization through the problem of human error. Because of our size, significant employee turnover, and remote workforce, we are a prime target for anybody looking to harm us. I strongly recommend implementing mandatory quarterly phishing simulations and a 48-hour access revocation policy for all departing employees. The cost of prevention is far less than the cost of recovery, and the time to act is now.

References

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